Per-Example Predictive Multiplicity in Hybrid Recurrent Architectures

Anonymous ACL submission

Abstract

Hybrid recurrent architectures that interleave attention with linear recurrence are increasingly deployed, yet how attention affects per-example behavior across backbones remains unquantified. Using the controlled model zoo of Wang et al. (2025)—three architectures, five attention ratios, two scales (340M/1.3B)—we evaluate 10 benchmarks and decompose per-example effects on four (LAMBADA, HellaSwag, PIQA, ARC-Easy). Per-example predictive multiplicity (Marx et al., 2020) varies across tasks ($\alpha = 0.14$–0.30) while pairwise Jaccard of helped/hurt sets remains uniformly low (≤ 0.10); multiplicity magnitude depends on task type, not attention dose. Aggregate convergence, by contrast, is task-specific: only LAMBADA shows monotonic variance reduction (79%), consistent with attention compensating for retrieval differences across recurrent backbones. On LAMBADA, aggregate variance decreases gradually while per-example overlap saturates at the first attention level—the two are decoupled.

1 Introduction

When attention layers are interleaved into recurrent architectures, they help and hurt almost entirely different examples—pairwise Jaccard overlap among three hybrid designs is at most 0.10, approximately twice the independence baseline. This per-example divergence is invisible in aggregate metrics: on LAMBADA, cross-architecture standard deviation drops 79% as attention increases (Bonferroni-adjusted $p = .100$; the BCa 95% CI excludes zero), but each architecture reaches similar accuracy by rescuing and disrupting different inputs. Practitioners increasingly choose between recurrent backbones based on aggregate benchmark scores; architectures that score identically in aggregate may thus carry hidden consequences for deployment reliability and ensemble construction.

This tension arises in the context of a rapid architectural shift toward hybrid recurrent-attention models, which interleave attention layers within linear recurrent backbones. Jamba (Lieber et al., 2024), Griffin (De et al., 2024), RWKV-6 (Peng et al., 2024), and RecurrentGemma (Botev et al., 2024) each combine recurrence with attention using different mixing strategies. These systems now serve as production-grade alternatives to pure Transformers, and the practical question of how much attention to include is typically answered by comparing aggregate benchmark numbers. Wang et al. (2025) provide a controlled model zoo spanning three recurrent architectures, five attention ratios, and two parameter scales, which enables systematic study of how attention allocation affects these hybrid designs.

Most evaluations of hybrid architectures compare aggregate benchmark scores and report that a given attention ratio achieves a certain accuracy averaged across tasks or averaged across architectures. This practice can mask substantial per-example behavioral diversity. Marx et al. (2020) formalized this risk as predictive multiplicity: a prediction problem admits competing models with conflicting predictions on individual inputs. D'Amour et al. (2022) documented a related phenomenon, underspecification, showing that models trained from the same pipeline with different random seeds can disagree on individual predictions despite equivalent aggregate performance. McCoy et al. (2020) demonstrated a concrete NLP instance: BERT models trained from different random seeds diverge on specific linguistic constructions. These studies examine variation from initialization or training noise within a single architecture. Whether predictive multiplicity extends to hybrid recurrent architectures, where the source of variation is the recurrent backbone itself, remains unquantified. The question matters because practitioners increasingly choose between recurrent backbones based on ag-

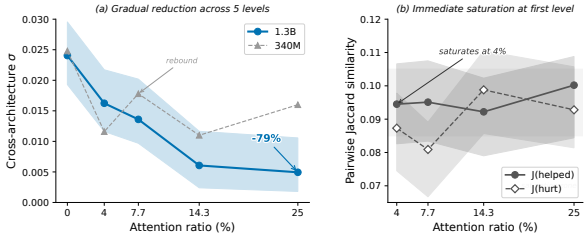


Figure 1: Aggregate convergence and per-example divergence are decoupled. (a) Cross-architecture standard deviation on LAMBADA decreases gradually with attention ratio at 1.3B (solid) but partially rebounds at 340M (dashed). (b) Pairwise Jaccard similarity of per-example helped/hurt sets saturates at the first attention level (4%) and remains flat.

gregate benchmarks; if architectures that score identically in aggregate behave differently on individual inputs, such choices carry hidden consequences for deployment reliability and ensemble construction.

We present the first systematic per-example analysis of how attention affects cross-architecture agreement in hybrid recurrent models. Using the model zoo of Wang et al. (2025), we compare three architectures (GatedDeltaNet, RetNet, HGRN2) at five attention ratios (0% to 25%) and two scales (340M and 1.3B parameters). These three architectures represent distinct approaches to linear recurrence: gated delta rules, multi-scale exponential decay, and gated state expansion. We evaluate all 30 configurations on 10 benchmarks spanning retrieval, comprehension, reasoning, and mixed task categories. On four benchmarks—LAMBADA ($N = 5{,}153$), HellaSwag ($N = 10{,}042$), PIQA ($N = 1{,}838$), and ARC-Easy ($N = 2{,}376$)—we conduct per-example decompositions, classifying each example as helped, hurt, or unchanged when attention is added, and measure cross-architecture overlap using Jaccard similarity. Figure 1 previews our central result: aggregate convergence and per-example divergence follow different trajectories as attention ratio increases.

Our analysis yields three findings. First, per-example predictive multiplicity varies in magnitude across tasks, forming a task-dependent spectrum. The ambiguity α (Marx et al., 2020) ranges from 0.14 (PIQA) through 0.15 (HellaSwag) and 0.21 (ARC-Easy) to 0.30 (LAMBADA), while pairwise Jaccard remains uniformly low (≤ 0.10, all $p < .01$). Within each benchmark, α is stable across attention ratios (variation < 0.03), but differs substantially across task types, ranging from 0.14 (reasoning) to 0.30 (retrieval). Second, attention-driven aggregate convergence is task-specific. Of 10 benchmarks, only LAMBADA exhibits monotonic reduction in cross-architecture variance ($\rho = -1.0$, BCa 95% CI $[-1.0, -0.7]$; cross-architecture σ drops 79%). LAMBADA (Paperno et al., 2016) is a pure context-retrieval task that requires predicting the final word of a passage from long-range dependencies; attention provides direct token-level access to the full context, which offers a natural explanation for why convergence occurs on this task and not others. Benchmarks that involve reasoning or comprehension show no convergence, with all $|\rho| \leq 0.7$ and $p \geq .233$. The Bonferroni-adjusted $p = .100$ does not survive strict correction at $\alpha = 0.05$; the confidence interval, which excludes zero, provides a complementary characterization of the effect. This convergence is scale-dependent: monotonic at 1.3B but non-monotonic at 340M (Appendix C). On LAMBADA, aggregate σ decreases gradually ($0.024 \rightarrow 0.005$) while per-example Jaccard saturates at 4% attention ($\sim$0.09)—the two quantities are decoupled. The experiments that follow begin with convergence (§4.1) because it establishes the aggregate backdrop against which per-example divergence (§4.2) and decoupling (§4.3) are measured.

Our contributions are as follows:

- We show that architectural variation—under matched training conditions—produces structured per-example predictive multiplicity with a task-dependent spectrum ($\alpha = 0.14$–0.30) across four benchmarks in three recurrent backbones, with magnitude independent of aggregate convergence (§4.2, §4.3, Table 2). Unlike seed-driven disagreement within a single architecture (McCoy et al., 2020; D'Amour et al., 2022), this multiplicity is not addressable by seed averaging: the source of variation is the recurrent backbone itself.

- We quantify task-specific convergence: only retrieval tasks (LAMBADA) show attention-driven cross-architecture convergence, and this pattern is scale-dependent (Appendix C; §4.1, Table 1, Figure 2).

2 Related Work

Hybrid Recurrent Architectures. The design space for combining attention with recurrence

spans layer-level interleaving (Jamba; Lieber et al., 2024), local attention pairing (Griffin, RecurrentGemma; De et al., 2024; Botev et al., 2024), matrix-valued recurrence (RWKV-6; Peng et al., 2024), within-head mixing (Hymba; Dong et al., 2024), and unbounded-context hybrid SSMs (Samba; Ren et al., 2024). Mamba-2 (Dao and Gu, 2024) establishes a duality between structured state spaces and attention; Mamba-3 (Lahoti et al., 2026) extends the design further, while Ali et al. (2025) reveal implicit attention within Mamba. Comparative evaluations confirm that hybrids outperform pure Transformers and SSMs at scale (Waleffe et al., 2024), and that even one attention layer enables content-based routing (Basu, 2026). Wang et al. (2025) provide the controlled model zoo we use, holding training conditions constant while varying architecture and attention ratio.

Linear Recurrent Models. Our three recurrent backbones represent distinct gating strategies within the linear recurrence family: RetNet (Sun et al., 2023) uses multi-scale exponential decay, HGRN2 (Qin et al., 2024) applies gated state expansion, and GatedDeltaNet (Yang et al., 2024) combines gated updates with delta rules. These descend from S4 (Gu et al., 2022) and Mamba (Gu and Dao, 2024), which introduced structured state spaces and input-dependent selection, respectively, and connect to linear attention (Katharopoulos et al., 2020) through the state-space–attention duality of Mamba-2.

Underspecification and Predictive Multiplicity. D'Amour et al. (2022) formalized underspecification: equivalent aggregate performance can mask per-prediction divergence. Concrete NLP instances include seed-dependent BERT disagreement (McCoy et al., 2020; Sellam et al., 2022) and the four-phase convergence pattern of autoregressive LMs (Fehlauer et al., 2025). Marx et al. (2020) introduced predictive multiplicity as a complementary lens, showing that near-optimal models produce conflicting per-example predictions; extensions address probabilistic classification (Watson-Daniels et al., 2023), Rashomon capacity (Hsu and Calmon, 2022), reconciliation (Du et al., 2025; Haghighat et al., 2026), and cross-model disagreement as a correctness signal (Gorbett and Jana, 2026). These studies attribute divergence to initialization noise or the near-optimal set within a single architecture. We show that architectural variation in hybrid recurrent models produces analogous per-example multiplicity under matched training conditions.

3 Method

We describe the model configurations, evaluation benchmarks, and analytical framework used to measure how attention affects cross-architecture agreement at both aggregate and per-example levels.

3.1 Model Zoo

We use the pretrained model zoo of Wang et al. (2025), which provides controlled comparisons across architectures and attention ratios. The zoo contains three linear recurrent architectures: GatedDeltaNet (Yang et al., 2024), RetNet (Sun et al., 2023), and HGRN2 (Qin et al., 2024). These three architectures represent distinct approaches to linear recurrence: GatedDeltaNet uses gated delta rules, RetNet uses multi-scale exponential decay, and HGRN2 uses gated linear recurrence with state expansion. Each architecture is trained at five attention ratios: 0% (pure recurrent), 4%, 7.7%, 14.3%, and 25%. We adopt the hk-1 notation from the original work, where one attention layer is interleaved per every k+1 total layers: h3-1 corresponds to 25% attention, h6-1 to 14.3%, h12-1 to 7.7%, and h24-1 to 4%. All configurations are available at two scales: 340M and 1.3B parameters. Within each scale, all models share identical training data, tokenizer, and hyperparameters; they differ only in architecture and attention configuration. This controlled design isolates the effect of attention from confounds such as training data or optimization differences. The zoo also includes a Transformer baseline (pure attention) at both scales.

3.2 Task Taxonomy

We evaluate on 10 benchmarks grouped into four categories defined by the primary cognitive demand of each task. This taxonomy was fixed prior to analysis (see Appendix A for the confirmatory/exploratory split), not derived post hoc from observed results. The categories test whether the effect of attention on cross-architecture variance depends on task type.

- **Retrieval**: LAMBADA (Paperno et al., 2016), which requires predicting a final word from long-range context without reasoning or world knowledge.
- **Comprehension**: BoolQ, a reading comprehension task with yes/no answers.

- **Reasoning**: HellaSwag, ARC-Challenge (Clark et al., 2018), WinoGrande, and PIQA. COPA ($n = 100$) serves as a supplementary reasoning benchmark; its small sample size renders α estimation unreliable (CI width 0.13–0.18; Appendix R).
- **Mixed**: ARC-Easy, MMLU, SciQ, and XStoryCloze, which combine retrieval, comprehension, and reasoning demands.

All evaluations use the lm-evaluation-harness framework (Gao et al., 2024) with default prompts; accuracy serves as the primary metric for all benchmarks.

3.3 Convergence Metrics

We quantify cross-architecture agreement as a function of attention ratio. For each benchmark and attention ratio, we compute the cross-architecture standard deviation σ over the three architectures' accuracy scores. To measure the relationship between attention and convergence, we compute the Spearman rank correlation ρ between attention ratio rank and σ across the five attention configurations ($n = 5$). A negative ρ indicates that higher attention ratios are associated with lower cross-architecture variance.

We also report the convergence ratio $\sigma_{\text{pure}}/\sigma_{\text{h3-1}}$, defined as the standard deviation under the pure-recurrent configuration divided by that under the highest attention ratio (25%). Values greater than 1 indicate that attention reduces cross-architecture variance.

3.4 Per-Example Decomposition

To move beyond aggregate metrics, we decompose performance changes at the individual-example level on four benchmarks spanning three task categories: LAMBADA (5,153 examples; retrieval), HellaSwag (10,042; reasoning), PIQA (1,838; reasoning, two-choice), and ARC-Easy (2,376; mixed). HellaSwag uses normalized accuracy (acc_norm) as the per-example correctness criterion. Supplementary analyses on BoolQ (3,270; comprehension) and WinoGrande (1,267; reasoning, two-choice) appear in Appendix O and P. For each architecture, we compare the pure-recurrent baseline against its h3-1 hybrid (25% attention), which maximizes the contrast between recurrent and hybrid modes; §4.2 confirms that α is stable across intermediate ratios (variation < 0.03). We classify each example into one of three categories:

- **Helped**: correct in the hybrid but incorrect in the pure model.
- **Hurt**: correct in the pure model but incorrect in the hybrid.
- **Unchanged**: both models agree (both correct or both incorrect).

The net effect for each architecture is $|\text{helped}| - |\text{hurt}|$. A positive net effect means attention produces more correct predictions than it removes; a negative net effect means the opposite.

To measure cross-architecture overlap in per-example effects, we compute the pairwise Jaccard similarity $J = |A \cap B|/|A \cup B|$ between the helped sets of two architectures and, separately, between their hurt sets. $J = 1$ indicates identical sets; $J = 0$ indicates no overlap. Low J values would indicate that different architectures are affected on different subsets of examples, which would constitute per-example divergence even when aggregate performance converges.

3.5 Statistical Methods

All statistical tests are two-tailed unless otherwise noted.

Permutation test. With only $n = 5$ attention ratios, we use an exact permutation test for ρ. All $5! = 120$ permutations of the attention ratio ranks are enumerated, and the two-tailed p-value is the fraction of permutations yielding $|\rho| \geq |\rho_{\text{obs}}|$.

Bootstrap confidence intervals. We construct 95% BCa (bias-corrected and accelerated) bootstrap confidence intervals for ρ using 10,000 resamples at the architecture level. With only $n = 3$ architectures, the resampling space is limited (at most 27 unique resamples before considering ratio permutations), so these intervals should be interpreted as heuristic indicators of effect direction rather than intervals with formal coverage guarantees. When the BCa correction degenerates at a boundary value, we report percentile confidence intervals instead. These intervals provide a complementary characterization of effect magnitude alongside the permutation test.

Multiple comparisons. We apply Bonferroni correction across six confirmatory benchmarks (Appendix A). The four remaining benchmarks are corrected separately across all 10 benchmarks and reported as exploratory.

Within- and cross-architecture tests. For within-architecture comparisons of per-example properties (e.g., token count of helped vs. hurt examples), we use the Mann-Whitney U test. For cross-architecture comparisons of per-example effect distributions, we use the Kruskal-Wallis test. Both are rank-based nonparametric tests, chosen because the per-example effect distributions are non-normal and contain ties.

Attention weight analysis. To probe whether attention mechanisms differ between helped and hurt examples, we extract attention weight matrices from each attention layer via forward hooks and compute two quantities: (1) cosine similarity between the mean attention distributions of two architectures at each layer, which measures cross-architecture agreement in attention allocation; and (2) Shannon entropy of the attention distribution for each example, which measures how concentrated or diffuse the attention pattern is. Details of the extraction procedure appear in Appendix F.

4 Experiments

We analyze how attention affects cross-architecture performance variance using models from the hybrid recurrent model zoo of Wang et al. (2025). Our analysis spans three architectures (GatedDeltaNet, RetNet, HGRN2), five attention ratios (0% pure recurrent, 4%, 7.7%, 14.3%, 25%), and two scales (340M and 1.3B parameters). In the hk-1 notation used throughout, one attention layer is interleaved per every k+1 layers; h3-1 denotes 25% attention. All models within each scale share identical training data and hyperparameters; they differ only in architecture and attention configuration. We evaluate on 10 benchmarks spanning retrieval, comprehension, reasoning, and mixed task categories, plus one supplementary benchmark (COPA, $n = 100$).

4.1 Task-Specific Convergence

Table 1 reports cross-architecture σ at each attention ratio for all benchmarks at 1.3B scale. LAMBADA is the only benchmark exhibiting monotonic convergence: $\rho = -1.0$ ($p = .017$; BCa 95% CI $[-1.0, -0.7]$; σ drops 79%; Bonferroni-adjusted $p = .100$). This is consistent with attention compensating for differences in how recurrent architectures retrieve from context (Arora et al., 2024a)—LAMBADA requires predicting a final word from long-range dependencies, and attention provides direct token-level access that narrows architectural

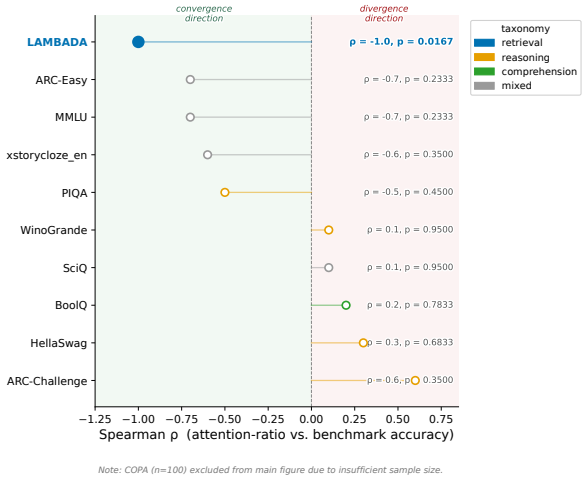


Figure 2: Spearman ρ between attention ratio and cross-architecture σ for each benchmark at 1.3B scale. LAMBADA ($\rho = -1.0$; BCa 95% CI $[-1.0, -0.7]$) is the only benchmark with a negative correlation whose confidence interval excludes zero. Benchmarks are grouped by task category.

differences in recall capacity (Arora et al., 2024b). No other benchmark shows significant convergence (all $|\rho| \leq 0.7$, $p \geq .233$; Figure 2). Full convergence discussion appears in Appendix B. Scale analysis (Appendix C) shows this convergence is monotonic at 1.3B but non-monotonic at 340M.

4.2 Per-Example Divergence

Do architectures converge by improving on the same examples? We compare each architecture's pure-recurrent baseline against its h3-1 hybrid on 5,153 LAMBADA examples, classifying each as helped (correct only in hybrid), hurt (correct only in pure), or unchanged. Although the decomposition is binary, we verify below that it captures genuine separation in the continuous confidence space (Cohen's $d = 0.73$–0.92 between helped and hurt on LAMBADA; cross-benchmark details in Appendix N.1).

Each architecture displays a distinct profile of per-example changes (Table 9). GDN loses a net 153 examples (462 helped, 615 hurt), RetNet loses 69 (426/495), and HGRN2 gains 153 (548/395). HGRN2 is the only architecture for which attention produces a net per-example benefit. Pairwise Jaccard similarity is low: mean $\bar{J} \leq 0.104$ for helped, ≤ 0.096 for hurt, indicating $\sim$90% non-overlap. Overlap significantly exceeds the independence baseline ($J_{\text{null}} \approx 0.05$, all $z > 5.9$, $p < .0001$; Appendix K), but normalizing by the theoretical upper bound $J_{\max} = \min(|A|, |B|)/\max(|A|, |B|)$, ob-

Table 1: Cross-architecture convergence across benchmarks (1.3B scale). ρ: Spearman correlation between attention ratio and cross-architecture σ across five configurations. p: exact permutation test (two-tailed, 120 permutations). σ values reported $\times 10^2$ (accuracy percentage points). Ratio: $\sigma_{\text{pure}}/\sigma_{\text{h3-1}}$. **Bold**: only benchmark with $\rho = -1.0$.

Benchmark	Category	ρ	p	σ_{pure}	$\sigma_{\text{h3-1}}$	Ratio
LAMBADA	**Retrieval**	$\mathbf{-1.0}$	**.017**	**2.41**	**0.49**	**4.87**
BoolQ	Comprehension	0.2	.783	1.63	2.36	0.69
ARC-C	Reasoning	0.6	.350	0.96	1.32	0.72
HellaSwag	Reasoning	0.4	.517	1.17	1.22	0.96
PIQA	Reasoning	-0.5	.450	0.54	0.34	1.59
WinoGrande	Reasoning	0.1	.950	0.67	0.78	0.86
ARC-E	Mixed	-0.7	.233	1.00	0.13	7.70
MMLU	Mixed	-0.7	.233	0.92	0.41	2.27
SciQ	Mixed	0.1	.950	0.87	0.74	1.19
XStoryCloze	Mixed	-0.6	.350	0.82	0.55	1.49
COPA†	Reasoning	-0.67	.267	2.63	0.47	5.57

†Supplementary ($n = 100$); excluded from multiple comparison correction.

Bonferroni-adjusted $p = 6 \times 2/120 = 0.100$ (exact); 0.102 arises from rounding raw p to .017 before multiplying.

served overlap reaches only 5–7% of the achievable range above chance (normalized ES 0.065 helped, 0.047 hurt; per-pair details in Appendix I), confirming that low Jaccard cannot be attributed to set-size constraints.

No surface-level feature (token count, word frequency, syntactic complexity) reliably separates helped from hurt examples. However, pure-model confidence on the target token is strongly predictive within each architecture: hurt examples have lower target-token probability among pure-correct examples ($d = 0.83$–0.92), and helped examples have higher probability among pure-wrong examples ($d = 0.73$–0.77); see Appendix N. A Kruskal-Wallis test confirms cross-architecture structure in both hurt ($p = .0005$) and helped ($p = .020$) distributions. The confidence margin thus identifies a per-example predictor of which examples will be affected: attention disrupts marginally correct predictions and rescues near-miss errors, but with architecture-specific thresholds that produce the observed cross-architecture divergence. The rescue pattern generalizes: across all three architectures on LAMBADA, HellaSwag, and ARC-Easy, helped examples have significantly higher pure-model target probability than stayed-wrong examples (9/9 tests, $d = 0.13$–0.60; Appendix N.1). The disruption pattern is weaker on ARC-Easy ($d = 0.13$–0.26), absent on HellaSwag under `acc_norm`, and absent on binary classification (Appendix O), likely reflecting structural properties of constrained-choice formats that limit the discriminative power of confidence margins. This structured per-example disagreement despite similar aggregate performance constitutes predictive multiplicity (Marx et al., 2020) arising from architectural variation rather than random initialization (D'Amour et al., 2022).

Multiplicity spectrum. Extending the per-example analysis to HellaSwag ($N = 10{,}042$) and PIQA ($N = 1{,}838$) reveals that predictive multiplicity varies systematically across tasks (Table 2). The ambiguity α (Marx et al., 2020) ranges from 0.14 (PIQA) through 0.15 (HellaSwag) and 0.21 (ARC-Easy) to 0.30 (LAMBADA), forming a task-dependent spectrum. Evaluation format does not drive multiplicity: HellaSwag (four-way, $\alpha = 0.15$) and PIQA (two-choice, $\alpha = 0.14$) produce nearly identical multiplicity despite different formats; among two-choice benchmarks alone, α spans $3\times$ from 0.14 (PIQA) to 0.41 (WinoGrande; Appendix P) to 0.45 (BoolQ; Appendix O), covering the full six-benchmark range within a single format. Bootstrap 95% CIs confirm the spectrum ordering: all adjacent-benchmark CIs are non-overlapping except PIQA and HellaSwag, whose CIs overlap substantially ($[.127, .159]$ vs. $[.141, .155]$; Appendix, Table 12). Moreover, α does not significantly correlate with task accuracy (Spearman $\rho = -0.60$, $p = .24$). Pairwise Jaccard remains uniformly low ($\bar{J} \leq 0.10$, all $p < .01$ across four benchmarks), indicating that architec-

Table 2: Predictive multiplicity at 1.3B. α: ambiguity at h3-1 (Marx et al., 2020). $\bar{J}$: mean pairwise Jaccard of helped (h) and hurt (u) sets, averaged across four hybrid attention ratios and three architecture pairs. ρ_α: Spearman ρ between attention ratio and α. Benchmarks ordered by α.

	N	α	$\bar{J}_h$	$\bar{J}_u$	ρ_α
PIQA	1838	.14	.091	.094	0.0
HellaSwag	10042	.15	.095	.085	0.1
ARC-Easy	2376	.21	.092	.096	-0.2
LAMBADA	5153	.30	.086	.091	-0.6

Within-benchmark α variation < 0.03 across ratios. BoolQ, WinoGrande: Appendix O, P.

tures affect different examples regardless of how many examples are affected. Within each benchmark, α is stable across attention ratios (variation < 0.03), but differs $2\times$ across task types (from reasoning to retrieval), confirming that multiplicity magnitude depends on task, not attention dose (cf. Gorbett and Jana, 2026). The near-identical α for PIQA (0.14) and HellaSwag (0.15) despite different evaluation formats suggests a shared reasoning-task floor for multiplicity, while retrieval (LAMBADA, 0.30) and mixed (ARC-Easy, 0.21) tasks produce distinctly higher values.

4.3 Decoupling

A key consequence of this multiplicity is that aggregate convergence and per-example divergence operate on different timescales as attention ratio increases (Figure 1).

Cross-architecture σ decreases gradually across the five attention ratios: $0.024 \rightarrow 0.016 \rightarrow 0.014 \rightarrow 0.006 \rightarrow 0.005$. In contrast, pairwise Jaccard similarity between architectures' per-example effects saturates at the first attention level (4%): approximately 0.095 for helped, with no further increase at higher ratios. Net asymmetry between architectures' per-example profiles increases from 0.013 at 4% attention to 0.040 at 25%. If convergence arose from attention uniformly rescuing the same difficult examples across architectures, Jaccard would increase in parallel with decreasing σ. The observed saturation of Jaccard at the first attention level indicates that each architecture's per-example response to attention is largely idiosyncratic.

This separation of timescales constitutes a decoupling: attention reduces the aggregate performance gap but does not homogenize per-example behavior. Bootstrap confidence intervals (10,000 resamples) confirm this statistically: the 95% CI for σ does not overlap between pure $[.019, .030]$ and h3-1 $[.002, .011]$, while the CIs for mean Jaccard overlap completely across all non-zero attention ratios (helped: $[.08, .11]$; hurt: $[.07, .11]$). Within-architecture Jaccard averages $\sim$0.33 (Appendix M), roughly $3\times$ the cross-architecture value ($\sim$0.10). Attributing this gap entirely to seed effects would require seed variation to exceed the effect of replacing 25% of model layers with attention—implausible though not directly testable with the available data.

Attention weight analysis (Appendix F) shows GDN and RetNet produce nearly identical attention distributions (cosine 0.995) yet differ in per-example effects, suggesting these originate in the recurrent layers. The four-benchmark spectrum (Table 2) confirms independence: LAMBADA ($\alpha = 0.30$) is the only benchmark showing convergence despite having the highest multiplicity, while PIQA ($\alpha = 0.14$) and HellaSwag ($\alpha = 0.15$) show no convergence with the lowest multiplicity. Cross-scale analysis on all four benchmarks reveals that the qualitative pattern of per-example multiplicity is scale-robust: pairwise Jaccard remains stable across scales ($\bar{J} = 0.08$–0.10 at 340M vs. 0.09–0.095 at 1.3B), and architecture-specific helped/hurt profiles persist. While α is systematically higher at 340M ($\Delta \leq 8$ pp), this inflation does not correlate with accuracy loss across benchmarks (HellaSwag shows the largest accuracy gap yet one of the smallest α increases), suggesting capacity-related but not accuracy-driven modulation. At 340M, aggregate convergence is absent on all benchmarks ($\rho = -0.50$, $p = .45$ for LAMBADA). This dissociation suggests that the *pattern* of per-example multiplicity—which architectures affect which examples—reflects structural differences in recurrent processing, while absolute α magnitude is partially modulated by model capacity (Appendix Q).

4.4 Robustness

Removing GDN weakens LAMBADA convergence from $\rho = -1.0$ to $\rho = -0.7$ ($p = .233$), partly because GDN's h3-1 uses 16 attention heads rather than 32 in all other configurations (Appendix E). The RetNet–HGRN2 pair preserves the decoupling: all hybrid $\sigma \leq 0.008$ while Jaccard saturates at 0.086–0.108. The aggregate convergence result is therefore partially dependent on GDN; the decoupling result is not. Full sensitivity analyses

appear in Appendix D.

5 Conclusion

We present the first systematic per-example analysis of attention effects in hybrid recurrent models. Per-example multiplicity spans a task-dependent spectrum ($\alpha = 0.14$–0.30) across four benchmarks, with Jaccard uniformly low (≤ 0.10) regardless of aggregate convergence, suggesting diverse hybrid ensembles may offer complementary coverage. Aggregate convergence, by contrast, is task-specific (only LAMBADA) and scale-dependent (monotonic at 1.3B, non-monotonic at 340M). For model selection, our results indicate that aggregate benchmark comparisons understate the behavioral differences between hybrid designs; per-example evaluation should inform deployment decisions where prediction consistency matters. While low Jaccard overlap is suggestive, ensemble gains require empirical validation beyond the scope of this observational study.

Limitations

Our analysis has several limitations.

Coverage. Per-example analysis covers four main benchmarks; BoolQ and WinoGrande are supplementary (Appendix O, P). The correlation between α and task accuracy is moderate but not significant ($\rho = -0.60$, $p = .24$); HellaSwag (60%, $\alpha = 0.15$) and PIQA (73%, $\alpha = 0.14$) show low α despite varied accuracy, while benchmarks near chance level (BoolQ, WinoGrande) show the highest α. Accuracy alone does not explain the spectrum, though we cannot fully rule out more complex interactions. We examine three recurrent architectures from one model zoo; the controlled design (shared training data, tokenizer, hyperparameters) isolates architectural effects but limits generalization to other hybrid families.

Methodological constraints. With only $n = 5$ attention ratios, the minimum achievable two-tailed p-value is approximately 0.008, contributing to the Bonferroni-adjusted $p = .100$ for LAMBADA not reaching conventional significance. Each zoo configuration corresponds to a single training run; we cannot fully disentangle architectural from seed-specific effects, though the $3\times$ Jaccard gap (§4.3) provides proxy evidence for architecture as the dominant factor. At 340M, α is systematically higher than at 1.3B (by up to 8.4pp on ARC-Easy), but pairwise Jaccard remains stable across scales ($\bar{J} = 0.08$–0.10), indicating that the qualitative pattern of per-example multiplicity is more robust than its absolute magnitude. Our attention weight analysis is descriptive, not causal; we compare only two scale points (340M, 1.3B)—a binary observation, not a scaling law.

Pre-training only. All models are evaluated after pre-training without fine-tuning; per-example effects and the decoupling pattern may not persist after task-specific adaptation.

Ethics Statement

This work analyzes publicly available pretrained language models from the model zoo of Wang et al. (2025). No new data collection, human subjects research, or model training is involved. We identify per-example behavioral differences across architectures, which may inform practitioners about the limitations of aggregate benchmark comparisons but does not introduce new risks.

References

Ameen Ali, Itamar Zimerman, and Lior Wolf. 2025. The hidden attention of mamba models. In *Proceedings of ACL*.

Simran Arora, Sabri Eyuboglu, Aman Timalsina, Isys Johnson, Michael Poli, James Zou, Atri Rudra, and Christopher Ré. 2024a. Zoology: Measuring and improving recall in efficient language models. In *Proceedings of ICLR*.

Simran Arora, Sabri Eyuboglu, Michael Zhang, Aman Timalsina, Frederic Alberti, James Bornschein, and 1 others. 2024b. Simple linear attention language models balance the recall-throughput tradeoff. In *Proceedings of ICML*.

Abhinaba Basu. 2026. When does content-based routing work? representation requirements for selective attention in hybrid sequence models. *arXiv preprint arXiv:2603.20997*.

Aleksandar Botev, Soham De, Samuel L. Smith, and 1 others. 2024. Recurrentgemma: Moving past transformers for efficient open language models. *arXiv preprint arXiv:2404.07839*.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*.

Alexander D'Amour, Katherine Heller, Dan Moldovan, and 1 others. 2022. Underspecification presents challenges for credibility in modern machine learning.

Journal of Machine Learning Research, 23(226):1–61.

Tri Dao and Albert Gu. 2024. Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality. In *Proceedings of ICML*.

Soham De, Samuel L. Smith, Anushan Fernando, and 1 others. 2024. Griffin: Mixing gated linear recurrences with local attention for efficient language models. *arXiv preprint arXiv:2402.19427*.

Xin Dong and 1 others. 2024. Hymba: A hybrid-head architecture for small language models. *arXiv preprint arXiv:2411.13676*.

Ally Yalei Du, Dung Daniel Ngo, and Zhiwei Steven Wu. 2025. Reconciling model multiplicity for downstream decision making. In *Proceedings of ICLR*.

Finlay Fehlauer, Kyle Mahowald, and Tiago Pimentel. 2025. Convergence and divergence of language models under different random seeds. In *Proceedings of EMNLP*, pages 32982–32991.

Leo Gao, Jonathan Tow, Baber Abbasi, Stella Biderman, and 1 others. 2024. A framework for few-shot language model evaluation. Zenodo.

Matt Gorbett and Suman Jana. 2026. Cross-model disagreement as a label-free correctness signal. *arXiv preprint arXiv:2603.25450*.

Albert Gu and Tri Dao. 2024. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*.

Albert Gu, Karan Goel, and Christopher Ré. 2022. Efficiently modeling long sequences with structured state spaces. In *Proceedings of ICLR*.

Parian Haghighat, Hadis Anahideh, and Cynthia Rudin. 2026. Resolving predictive multiplicity for the rashomon set. *arXiv preprint arXiv:2601.09071*.

Hsiang Hsu and Flavio P. Calmon. 2022. Rashomon capacity: A metric for predictive multiplicity in classification. In *Advances in Neural Information Processing Systems*.

Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. 2020. Transformers are RNNs: Fast autoregressive transformers with linear attention. In *Proceedings of ICML*.

Aakash Lahoti, Kevin Y. Li, Berlin Chen, Caitlin Wang, Aviv Bick, J. Zico Kolter, Tri Dao, and Albert Gu. 2026. Mamba-3: Improved sequence modeling using state space principles. In *Proceedings of ICLR*.

Opher Lieber, Barak Lenz, Hofit Bata, and 1 others. 2024. Jamba: A hybrid transformer-mamba language model. *arXiv preprint arXiv:2403.19887*.

Charles Marx, Flavio P. Calmon, and Berk Ustun. 2020. Predictive multiplicity in classification. In *Proceedings of ICML*.

R. Thomas McCoy, Junghyun Min, and Tal Linzen. 2020. BERTs of a feather do not generalize together: Large variability in generalization across models with similar test set performance. In *Proceedings of NeurIPS*.

Denis Paperno, Germán Kruszewski, Angeliki Lazaridou, and 1 others. 2016. The LAMBADA dataset: Word prediction requiring a broad discourse context. In *Proceedings of ACL*.

Bo Peng and 1 others. 2024. Eagle and finch: RWKV with matrix-valued states and dynamic recurrence. *arXiv preprint arXiv:2404.05892*.

Zhen Qin and 1 others. 2024. HGRN2: Gated linear RNNs with state expansion. *arXiv preprint arXiv:2404.07904*.

Liliang Ren, Yang Liu, Yadong Yan, Yi Zhu, and 1 others. 2024. Samba: Simple hybrid state space models for efficient unlimited context language modeling. *arXiv preprint arXiv:2406.07522*.

Thibault Sellam, Steve Yadlowsky, Ian Tenney, Jason Wei, Naomi Saphra, Alexander D'Amour, Tal Linzen, Jasmijn Bastings, Iulia Raluca Turc, Jacob Eisenstein, Dipanjan Das, and Ellie Pavlick. 2022. The MultiBERTs: BERT reproductions for robustness analysis. In *Proceedings of ICLR*.

Yutao Sun, Li Dong, Shaohan Huang, Shuming Ma, Yuqing Xia, Jilong Xue, Jianyong Wang, and Furu Wei. 2023. Retentive network: A successor to transformer for large language models. *arXiv preprint arXiv:2307.08621*.

Roger Waleffe, Wonmin Byeon, Duncan Riach, Brandon Norick, and 1 others. 2024. An empirical study of mamba-based language models. *arXiv preprint arXiv:2406.07887*.

Dustin Wang, Rui-Jie Zhu, Angelos Katharopoulos, and Albert Gu. 2025. A systematic analysis of hybrid linear attention. *arXiv preprint arXiv:2507.06457*.

Jamelle Watson-Daniels, David C. Parkes, and Berk Ustun. 2023. Predictive multiplicity in probabilistic classification. In *Proceedings of AAAI*, volume 37, pages 10306–10314.

Songlin Yang, Jan Kautz, and Ali Hatamizadeh. 2024. Gated delta networks: Improving mamba2 with delta rule. *arXiv preprint arXiv:2412.06464*.

A Confirmatory and Exploratory Analyses

The task taxonomy used in §3.2 was defined prior to running any convergence analyses. Each benchmark was assigned to exactly one of four categories based on the primary cognitive demand of the task:

- **Retrieval**: tasks requiring prediction from long-range context without multi-step reasoning (LAMBADA).

- **Comprehension**: reading comprehension with explicit passage support (BoolQ).
- **Reasoning**: tasks requiring commonsense inference or multi-step reasoning (HellaSwag, ARC-Challenge, WinoGrande, PIQA).
- **Mixed**: tasks combining retrieval, comprehension, and reasoning demands (ARC-Easy, MMLU, SciQ, XStoryCloze).

Confirmatory benchmark set. Six benchmarks were designated as confirmatory for Bonferroni correction: LAMBADA, HellaSwag, ARC-Challenge, WinoGrande, ARC-Easy, and MMLU. These were selected as representative of retrieval, reasoning, and mixed categories before any convergence analysis was performed. The taxonomy and benchmark selection were fixed before analysis; we describe this as a confirmatory/exploratory split rather than a formal pre-registration, since no external timestamp exists. The Bonferroni-adjusted significance threshold is $\alpha/6 = 0.0083$.

Exploratory benchmarks. Four additional benchmarks (BoolQ, PIQA, SciQ, XStoryCloze) were added as exploratory tests. These are corrected separately across all 10 benchmarks (adjusted threshold $\alpha/10 = 0.005$) and reported as exploratory throughout the paper.

Supplementary benchmark. COPA ($n = 100$) is treated as supplementary due to its small sample size. It is excluded from both correction families and reported separately with the † annotation in Table 1.

B Convergence Detail

Beyond LAMBADA ($\rho = -1.0$), two benchmarks show moderate negative correlations that do not reach significance: ARC-Easy and MMLU (both $\rho = -0.7$, $p = .233$). Both are classified as "Mixed" tasks combining retrieval with reasoning demands. The sensitivity analysis (Appendix D) shows that MMLU's correlation reverses upon GDN exclusion ($\rho = +0.6$), indicating dependence on a single architecture; ARC-Easy's correlation is more stable. COPA ($n = 100$; supplementary) shows $\rho = -0.67$ with a large σ ratio (5.57), consistent with LAMBADA's pattern but unreliable due to small sample size.

Several benchmarks show positive or near-zero ρ (BoolQ: $+0.2$; ARC-C: $+0.6$; HellaSwag: $+0.4$; WinoGrande: $+0.1$; SciQ: $+0.1$), indicating that attention does not systematically reduce cross-architecture variance on reasoning or comprehension tasks. The contrast with LAMBADA is consistent with attention's convergence benefit being specific to tasks requiring direct token-level recall (Arora et al., 2024a,b) rather than multi-step inference.

Regression to the mean. The direction of LAMBADA's aggregate convergence is consistent with regression to the mean: GDN (highest pure accuracy, 0.543) declines to 0.513 under h3-1, while HGRN2 (lowest, 0.484) rises to 0.514, yielding a $5\times$ σ compression ($0.024 \rightarrow 0.005$). However, three observations argue against a purely statistical artifact. First, the $5\times$ reduction in cross-architecture σ far exceeds typical regression amplitudes. Second, the direction is entirely determined by architecture identity, not random fluctuation: HGRN2 accuracy increases monotonically across all attention ratios, while GDN decreases monotonically—a pattern inconsistent with noise-driven mean reversion. Third, per-example analysis (§4.2) reveals architecture-specific mechanisms: GDN's net -153 examples and HGRN2's net $+153$ examples reflect distinct helped/hurt profiles with low cross-architecture Jaccard overlap, indicating that the aggregate convergence arises from structured, architecture-dependent per-example changes rather than symmetric regression.

C Scale-Dependent Convergence

At 1.3B parameters, LAMBADA convergence is monotonic: σ decreases at every attention level from 0% through 25% ($\rho = -1.0$). At 340M, convergence is partial: σ reaches a maximum reduction of 56% at intermediate attention ratios but rebounds at h3-1, yielding a net endpoint reduction of 36%. The trajectory is non-monotonic, and the highest-attention model does not achieve the lowest cross-architecture variance. One possible interpretation is that at smaller scale, the capacity overhead of attention layers creates interference that offsets the retrieval benefit.

This contrast is a preliminary observation from two data points (not a scaling law). Despite the aggregate divergence at 340M, mean pairwise Jaccard is similar at both scales ($J \approx 0.09$), extending the decoupling observation (§4.3) to the scale dimension.

D Sensitivity Analyses

D.1 Bootstrap Confidence Intervals

We construct 95% bootstrap confidence intervals for the Spearman ρ between attention ratio and cross-architecture σ on LAMBADA using 10,000 resamples at the architecture level ($n = 3$ architectures). Of these resamples, 11.25% are degenerate: they consist of same-architecture triplicates, which produce a constant σ vector and therefore an undefined ρ (division by zero in the rank correlation formula). These are excluded from the confidence interval computation because they carry no information about the correlation, not because they represent $\rho = 0$.

The BCa (bias-corrected and accelerated) correction degenerates at the $\rho = -1.0$ boundary because the acceleration parameter requires finite jackknife variance, which collapses when the point estimate is at the distributional limit. We therefore report percentile confidence intervals instead. From the 88.75% informative resamples, the 95% percentile CI is $[-1.0, -0.7]$, which excludes zero.

D.2 GDN Exclusion Sensitivity

The GDN h3-1 configuration uses 16 attention heads rather than 32 (see Appendix E). To assess the influence of this confound, we recompute all convergence correlations after excluding GDN, leaving two architectures (RetNet, HGRN2) at five attention ratios.

Table 3 reports the results for benchmarks where the correlation changes by $|\Delta\rho| \geq 0.3$. LAMBADA weakens from $\rho = -1.0$ to $\rho = -0.7$ ($p = .233$), consistent with GDN contributing to the convergence signal. MMLU reverses from $\rho = -0.7$ to $\rho = +0.6$, confirming that the moderate negative correlation on MMLU depends on GDN's inclusion and should not be interpreted as evidence of convergence.

Table 3: Effect of GDN exclusion on Spearman ρ (attention ratio vs. cross-architecture σ) at 1.3B scale. Only benchmarks with $|\Delta\rho| \geq 0.3$ shown.

Benchmark	ρ_{all}	$\rho_{\text{no-GDN}}$	$\Delta\rho$	$p_{\text{no-GDN}}$
LAMBADA	-1.0	-0.7	$+0.3$	.233
MMLU	-0.7	$+0.6$	$+1.3$	.350

E GDN Attention Head Confound

In the Wang et al. (2025) model zoo, the GDN h3-1 configuration at 1.3B scale uses 16 attention heads per attention layer, while all other hybrid configurations (across GDN, RetNet, and HGRN2) use 32 attention heads. This is a design choice in the original model zoo, not an error. The reduced head count changes the per-head dimension and may affect the attention capacity of GDN h3-1 relative to other configurations.

This confound is relevant because the convergence analysis measures cross-architecture σ at each attention ratio. At the h3-1 ratio (25% attention), GDN's attention layers have a different internal structure than those of RetNet and HGRN2, which could contribute to variance reduction (or inflation) for reasons unrelated to the recurrent–attention interaction under study.

The sensitivity analysis in Appendix D shows that removing GDN weakens the LAMBADA convergence correlation from $\rho = -1.0$ to $\rho = -0.7$. This weakening is partially attributable to the reduced sample size ($n = 2$ architectures) and partially to the confound itself. We cannot separate these two effects with the available data.

F Attention Weight Analysis

We extract attention weight matrices from the 1.3B h3-1 models of all three architectures using forward hooks on each attention layer. While Ali et al. (2025) show that SSM layers learn implicit attention patterns, our analysis focuses on the explicit attention layers interleaved within the hybrid architecture. For each layer, we compute the mean attention distribution across all 5,153 LAMBADA examples and measure cross-architecture similarity via cosine similarity.

F.1 Cross-Architecture Cosine Similarity

Table 4 reports cosine similarity between mean attention distributions at selected layers and averaged across all layers.

Table 4: Cosine similarity between mean attention weight distributions of architecture pairs at selected layers (1.3B h3-1 models on LAMBADA).

Layer	GDN–RetNet	GDN–HGRN2	HGRN2–RetNet
L2 (early)	0.995	0.244	0.196
L20 (deep)	0.986	0.942	0.943
Mean (all layers)	0.905	0.766	0.698

At layer 2, GDN and RetNet produce nearly identical attention distributions (cosine 0.995), while HGRN2 diverges from both (cosine < 0.25). By layer 20, all three architectures converge to similar

attention patterns (cosine > 0.94). The layer-wise trajectory suggests that early-layer attention patterns are shaped by the preceding recurrent layers (which differ across architectures), while deeper layers converge regardless of the recurrent substrate.

F.2 Helped vs. Hurt Attention Entropy

To test whether the attention mechanism itself differs between helped and hurt examples, we compute the Shannon entropy of the attention distribution for each example at each attention layer and compare the mean entropy of helped examples against hurt examples within each architecture.

Table 5: Mean attention entropy difference (helped − hurt) averaged across attention layers (1.3B h3-1 models on LAMBADA). Values near zero indicate that the attention mechanism does not discriminate between helped and hurt examples.

Architecture	ΔH (helped − hurt)
GDN	-0.0009
RetNet	$+0.0179$
HGRN2	$+0.0020$

GDN and HGRN2 show negligible entropy differences ($|\Delta H| \leq 0.002$). RetNet's $\Delta H = +0.018$ is small in absolute terms but slightly larger; helped examples receive marginally more diffuse attention in RetNet. Across all three architectures, the differences are insufficient to explain the per-example divergence, consistent with per-example effects not being explained by attention weight patterns and likely originating in the recurrent layers.

F.3 Last-5 Attention Mass

We also measure the fraction of attention mass allocated to the last five tokens, which captures whether the attention mechanism concentrates differently on recent context for helped vs. hurt examples.

Table 6: Mean attention mass on last-5 tokens for helped and hurt examples, averaged across attention layers (1.3B h3-1 models on LAMBADA).

Architecture	Helped	Hurt
GDN	0.0834	0.0831
RetNet	0.1578	0.1591
HGRN2	0.0453	0.0467

No architecture shows a meaningful difference in last-5 attention mass between helped and hurt examples (Table 6). This further supports the conclusion that the per-example divergence is not driven by differential attention allocation.

G Cross-Scale Per-Example Comparisons

G.1 Within-Scale Pairwise Jaccard

Table 7 reports the within-scale pairwise Jaccard similarity between architectures' helped and hurt sets at both scales. Mean values are similar across scales (340M: 0.094/0.090; 1.3B: 0.100/0.093), confirming that per-example architecture specificity is approximately scale-robust.

Table 7: Within-scale pairwise Jaccard similarity of per-example effects (pure → h3-1) on LAMBADA. Each row compares the helped (or hurt) sets of two architectures at the same scale.

Scale	Pair	J_{helped}	J_{hurt}
340M	GDN–RetNet	0.092	0.109
	GDN–HGRN2	0.084	0.075
	RetNet–HGRN2	0.108	0.085
	Mean	*0.094*	*0.090*
1.3B	GDN–RetNet	0.092	0.103
	GDN–HGRN2	0.100	0.089
	RetNet–HGRN2	0.108	0.087
	Mean	*0.100*	*0.093*

G.2 Cross-Scale Pairwise Jaccard

For each architecture, we compare the helped and hurt sets of the pure → h3-1 transition at 340M and 1.3B scale on LAMBADA. The comparison uses the same classification scheme as §4.2: an example is "helped" if it is correct only in the hybrid variant and "hurt" if correct only in the pure baseline. All pairwise Jaccard values fall below 0.10, indicating that scale produces per-example predictive multiplicity analogous to the cross-architecture effect reported in §4.2.

Table 8: Cross-scale pairwise Jaccard similarity of per-example effects (pure → h3-1) between 340M and 1.3B on LAMBADA. All values below 0.10.

Architecture	Scale Pair	J_{helped}	J_{hurt}
GDN	340M ↔ 1.3B	0.047	0.087
RetNet	340M ↔ 1.3B	0.059	0.073
HGRN2	340M ↔ 1.3B	0.089	0.055
Mean		0.065	0.072

H Per-Example Effect Counts

Table 9 reports the full per-example effect counts for each architecture at 1.3B scale.

Table 9: Per-example effects of attention (pure → h3-1) on 5,153 LAMBADA examples at 1.3B scale. $\bar{J}$: mean pairwise Jaccard similarity with other architectures across helped and hurt sets.

Architecture	Helped	Hurt	Net	$\bar{J}$
GDN	462	615	−153	.096
RetNet	426	495	−69	.100
HGRN2	548	395	+153	.104

I Normalized Jaccard Effect Size

To assess whether low observed Jaccard reflects a floor effect imposed by unequal set sizes, we compute the theoretical upper bound $J_{\max} = \min(|A|, |B|)/\max(|A|, |B|)$ and the normalized effect size $(J_{\text{obs}} - J_{\text{null}})/(J_{\max} - J_{\text{null}})$, which measures observed overlap as a fraction of the achievable range above chance. Table 10 reports per-pair results.

Table 10: Normalized Jaccard effect size for each architecture pair (1.3B h3-1 on LAMBADA). $J_{\max}$: theoretical upper bound given set sizes. Normalized ES: fraction of achievable overlap above chance.

Category	Pair	J_{obs}	$J_{\max}$	J_{null}	Norm. ES
Helped	GDN–RetNet	.086	.922	.048	.044
	GDN–HGRN2	.099	.843	.050	.062
	RetNet–HGRN2	.108	.777	.042	.090
	Mean	*.098*	*.847*	*.047*	*.065*
Hurt	GDN–RetNet	.092	.805	.056	.048
	GDN–HGRN2	.070	.642	.043	.045
	RetNet–HGRN2	.087	.798	.050	.049
	Mean	*.083*	*.748*	*.050*	*.047*

All normalized effect sizes fall below 0.10, confirming that observed overlap reaches only 5–7% of the theoretical maximum. The low Jaccard values cannot be attributed to set-size constraints.

J ARC-Easy Per-Example Analysis

Table 11 reports per-example effect counts on ARC-Easy ($N = 2{,}376$) at 1.3B scale for each attention ratio. Unlike LAMBADA, ARC-Easy shows no aggregate convergence ($\rho = -0.20$, $p = .783$). HGRN2 is net positive at all ratios; GDN and RetNet are mixed.

Table 12 reports the ambiguity α (Marx et al., 2020) at each attention ratio for all four main benchmarks. Within-benchmark ranges are 0.004 (HellaSwag), 0.011 (PIQA), 0.025 (ARC-Easy), and 0.022 (LAMBADA), all below the 0.03 threshold reported in §4.2.

Table 11: Per-example effects on ARC-Easy at 1.3B scale. Net = helped − hurt.

Architecture	Ratio	Helped	Hurt	Net
GDN	h24-1	178	185	−7
	h12-1	176	206	−30
	h6-1	170	157	+13
	h3-1	171	180	−9
RetNet	h24-1	158	168	−10
	h12-1	163	146	+17
	h6-1	159	147	+12
	h3-1	144	184	−40
HGRN2	h24-1	160	119	+41
	h12-1	159	111	+48
	h6-1	185	118	+67
	h3-1	164	146	+18

Table 12: Ambiguity α across attention ratios (1.3B scale). 95% CI: BCa bootstrap (10,000 resamples) at the pure-recurrent configuration.

Benchmark	Pure	h24-1	h12-1	h6-1	h3-1	95% CI (pure)
PIQA	.143	.138	.132	.143	.140	[.127, .159]
HellaSwag	.148	.150	.148	.146	.150	[.141, .155]
ARC-Easy	.205	.226	.223	.201	.210	[.189, .221]
LAMBADA	.307	.300	.316	.294	.296	[.294, .319]

K Jaccard Permutation Test

To assess whether the observed pairwise Jaccard similarities exceed what would be expected under independence, we perform a permutation test. For each architecture pair at 1.3B h3-1, we hold the helped (or hurt) set sizes fixed and randomly assign example indices to each set (10,000 permutations, seed 42). Table 13 reports the observed J, the null baseline J_{null} (mean over permutations), and the z-score.

All six comparisons yield $z > 5.9$ ($p < .0001$). The null baseline averages $J_{\text{null}} \approx 0.049$; observed values are approximately twice this level. The overlap between architectures' per-example effect sets is therefore structured rather than a product of random set overlap given the set sizes.

L Transformer Entrainment Comparison

To test whether attention layers make hybrid models' per-example behavior converge toward a pure

Table 13: Permutation test for pairwise Jaccard similarity (1.3B h3-1 on LAMBADA, 10,000 permutations). All observed values significantly exceed the independence baseline.

Category	Pair	J_{obs}	J_{null}	z	p
Helped	GDN–HGRN2	.100	.051	7.11	$<.0001$
	GDN–RetNet	.092	.045	6.75	$<.0001$
	HGRN2–RetNet	.108	.049	8.61	$<.0001$
Hurt	GDN–HGRN2	.088	.049	5.91	$<.0001$
	GDN–RetNet	.103	.056	6.85	$<.0001$
	HGRN2–RetNet	.087	.045	6.05	$<.0001$

Transformer, we compare the correct-example sets of each architecture's pure and h3-1 variants against a 1.3B Transformer baseline (accuracy 48.1%) on LAMBADA. Table 14 reports the Jaccard similarity between each model's correct set and the Transformer's correct set.

Table 14: Jaccard similarity between each model's correct set and the Transformer correct set on LAMBADA (1.3B scale). All three h3-1 hybrids are more similar to the Transformer than their pure counterparts.

Architecture	$J_{\text{pure,T}}$	$J_{\text{h3-1,T}}$	Δ
GDN	0.645	0.649	$+0.005$
RetNet	0.679	0.694	$+0.015$
HGRN2	0.672	0.701	$+0.029$

All three architectures show positive Δ: the h3-1 correct set is more similar to the Transformer correct set than the pure correct set is. The effect is strongest for HGRN2 ($+0.029$) and weakest for GDN ($+0.005$). This ordering is consistent with HGRN2 being the only architecture with a positive net per-example effect from attention (Table 9). The Transformer itself achieves the lowest accuracy among all models, so the convergence is not explained by hybrid models degrading toward Transformer-level performance; rather, adding attention shifts which specific examples the hybrid gets correct toward the Transformer's pattern.

M Within-Architecture Per-Example Stability

Per-example effect sets exhibit a three-tier Jaccard hierarchy on LAMBADA (1.3B scale). Under independence (permutation baseline), pairwise Jaccard is $J \approx 0.05$. Cross-architecture comparisons (different architectures at the same attention ratio) yield $J \approx 0.10$, approximately $2\times$ baseline. Within-architecture comparisons (same architecture across different attention ratios) yield $J \approx 0.33$, approximately $7\times$ baseline and $3\times$ the cross-architecture level. Architecture identity therefore has approximately three times the influence of attention dose on per-example effects.

Table 15 reports mean within-architecture Jaccard for each architecture. HGRN2 shows the highest helped overlap and lowest hurt overlap, consistent with its monotonically increasing helped count across ratios ($479 \rightarrow 548$).

Table 15: Mean within-architecture pairwise Jaccard similarity across attention ratios (1.3B, LAMBADA). Each mean is computed over all six ratio pairs per architecture.

Architecture	$\bar{J}_{\text{helped}}$	$\bar{J}_{\text{hurt}}$
GDN	0.338	0.319
RetNet	0.333	0.321
HGRN2	0.373	0.291
Overall mean	*0.348*	*0.310*

Table 16 reports the full pairwise Jaccard matrix. Attention ratios are denoted by their hk-1 labels: h24-1 (4%), h12-1 (7.7%), h6-1 (14.3%), h3-1 (25%).

Table 16: Within-architecture pairwise Jaccard similarity between helped and hurt sets at different attention ratios (1.3B, LAMBADA). Each cell compares the per-example effect sets of two attention configurations within the same architecture.

		Ratio pair					
Arch	Cat	h24–h12	h24–h6	h24–h3	h12–h6	h12–h3	h6–h3
GDN	Helped	.345	.335	.311	.344	.341	.352
	Hurt	.310	.328	.320	.331	.281	.345
RetNet	Helped	.354	.314	.334	.322	.330	.345
	Hurt	.316	.334	.307	.301	.324	.343
HGRN2	Helped	.359	.369	.375	.368	.378	.388
	Hurt	.292	.289	.298	.273	.309	.287

Adjacent ratio pairs (e.g., h24-1 vs. h12-1) do not exhibit systematically higher Jaccard than non-adjacent pairs (e.g., h24-1 vs. h3-1). Across all architectures, the mean Jaccard for adjacent pairs is 0.353 (helped) and 0.311 (hurt), compared with 0.343 (helped) and 0.310 (hurt) for non-adjacent pairs. The negligible difference indicates that the mapping from attention dose to per-example effects is non-smooth: doubling the attention ratio does not produce a proportionally more similar set of affected examples.

N Target-Token Confidence Margin

We analyze whether pure-model confidence on the target token predicts whether attention will help or hurt on a given example. We define confidence as $|\log p_{\text{pure}}(y^*)|$, where y^* is the target token: lower values indicate higher probability (near-correct). We initially tested raw perplexity as a predictor of helped/hurt status, but found that within-category controls (C1/C2) are necessary to isolate attention effects from the definitional confound between pure-correct and pure-wrong populations.

Table 17 compares hurt versus stayed-right samples (both pure-correct): hurt samples have significantly lower target-token probability (higher $|\log p|$), confirming that weakly-correct predictions are vulnerable to attention disruption. Table 18 compares helped versus stayed-wrong samples (both pure-wrong): helped samples have higher target-token probability (lower $|\log p|$), confirming that near-miss errors are rescuable. This pattern is consistent across all three architectures ($|d| = 0.73$–0.92, all $p < 10^{-54}$) but does not alter the cross-architecture Jaccard findings—within-architecture predictability coexists with cross-architecture divergence.

Table 17: Control 1: Pure-correct samples—hurt vs. stayed-right. Higher mean $|\log p|$ indicates lower target-token probability (less confident correct).

Architecture	n_{hurt}	n_{stay}	Hurt mean	Stay mean	Cohen's d
GDN (h12-1)	519	2281	1.049	0.576	0.87
RetNet (h3-1)	497	2169	1.293	0.785	0.83
HGRN2 (h3-1)	392	2102	1.481	0.899	0.92

Table 18: Control 2: Pure-wrong samples—helped vs. stayed-wrong. Lower mean $|\log p|$ indicates higher target-token probability (near-miss error).

Architecture	n_{help}	n_{stay}	Help mean	Stay mean	Cohen's d
GDN (h12-1)	480	1873	2.459	4.189	−0.75
RetNet (h3-1)	420	2067	2.478	4.103	−0.73
HGRN2 (h3-1)	548	2111	2.500	4.112	−0.77

N.1 Cross-Benchmark Extension

The confidence margin pattern extends beyond LAMBADA to multiple-choice benchmarks, though with smaller effect sizes. Table 19 reports the rescue analysis (helped vs. stayed-wrong) on HellaSwag and ARC-Easy. All nine comparisons (three benchmarks × three architectures) are significant, confirming that attention consistently rescues near-miss errors regardless of task type. Effect sizes range from small on HellaSwag ($d = 0.13$–0.17) to medium on ARC-Easy ($d = 0.51$–0.60), compared to large on LAMBADA ($d = 0.73$–0.77); the attenuation on HellaSwag reflects noisier confidence proxies under length-normalized scoring. Table 20 reports the disruption analysis (hurt vs. stayed-right); two of six comparisons are significant (all on ARC-Easy) with small effect sizes ($d = 0.13$–0.26), indicating that the disruption signal is weaker on multiple-choice tasks and absent on HellaSwag under `acc_norm`. The asymmetry between rescue and disruption reflects task structure: near-miss errors are an intrinsic sample property (the model almost produced the correct answer regardless of format), while borderline correctness in multiple-choice settings depends on relative option ranking rather than absolute confidence, reducing the raw log-likelihood margin's predictive power for vulnerability.

Table 19: Cross-benchmark rescue analysis (Control 2): helped vs. stayed-wrong on HellaSwag and ARC-Easy (1.3B h3-1).

Benchmark	Architecture	Cohen's d	p
HellaSwag	GDN	−0.166	6.1×10^{-4}
	RetNet	−0.132	1.4×10^{-2}
	HGRN2	−0.164	1.4×10^{-3}
ARC-Easy	GDN	−0.60	1.0×10^{-13}
	RetNet	−0.51	5.3×10^{-10}
	HGRN2	−0.53	5.4×10^{-10}

Table 20: Cross-benchmark disruption analysis (Control 1): hurt vs. stayed-right on HellaSwag and ARC-Easy (1.3B h3-1).

Benchmark	Architecture	Cohen's d	p	Sig.?
HellaSwag	GDN	0.009	.999	No
	RetNet	−0.035	.453	No
	HGRN2	−0.062	.274	No
ARC-Easy	GDN	0.26	3.2×10^{-4}	Yes
	RetNet	0.22	2.8×10^{-3}	Yes
	HGRN2	0.13	.145	No

O BoolQ Supplementary Analysis

BoolQ ($N = 3{,}270$; comprehension, binary yes/no classification) shows the highest ambiguity among the benchmarks we analyzed ($\alpha \approx 0.45$), but the Jaccard significance pattern is mixed: only 12 of 24 architecture × attention-ratio pair combinations yield $p < .01$. In particular, the RetNet–HGRN2 pair produces 0/8 significant Jaccard values, with

several normalized effect sizes below zero, suggesting anti-correlated per-example effects for this pair.

Binary classification baseline. BoolQ is binary classification (two answer choices), whereas LAMBADA, HellaSwag, and ARC-Easy involve open vocabulary prediction or four-way multiple choice. Under a binary task, the null Jaccard baseline J_{null} is higher because chance overlap between helped/hurt sets is larger when there are fewer ways to be wrong. This elevates the significance threshold: observed Jaccard must be substantially higher to exceed the null, which explains why 12/24 pairs fail to reach significance despite $\bar{J}_h = 0.103$ and $\bar{J}_u = 0.098$—values comparable to the four main benchmarks.

Per-example summary. Table 21 reports the key metrics. The high α reflects that BoolQ examples are more sensitive to architectural variation in the presence of attention: approximately half of all examples change correctness status for at least one architecture. However, the mixed Jaccard significance prevents us from confidently concluding that the cross-architecture overlap structure matches the pattern observed on the four main benchmarks.

Table 21: BoolQ predictive multiplicity at 1.3B (h3-1).

	N	α	$\bar{J}_h$	$\bar{J}_u$	Sig. pairs	ρ_α
BoolQ	3270	.45	.103	.098	12/24	+0.1

RetNet–HGRN2: 0/8 significant; multiple normalized ES < 0.

The significant Jaccard overlap concentrates in GDN–RetNet pairs (8/8 significant), while RetNet–HGRN2 shows none (0/8). This pair-specificity does not appear on the four main benchmarks, where all architecture pairs show significant overlap; it may reflect a BoolQ-specific interaction between binary classification structure and particular recurrent architectures rather than a general property of architecture pairs.

Confidence margin. Confidence margin analysis on BoolQ shows the rescue pattern (C2) partially holds (2/3 architectures significant, $d = 0.16$–0.36) but the disruption pattern (C1) is absent (0/3, $|d| \leq 0.10$), consistent with compressed log-likelihood ranges in binary classification reducing the predictive power of confidence margins. This extends the gradient observed across LAMBADA, ARC-Easy, and HellaSwag: C1 effect sizes decrease from open-ended prediction ($d \approx 0.9$, LAMBADA) through ARC-Easy ($d \approx 0.1$–0.3) to HellaSwag ($d \approx 0$ under `acc_norm`) and binary classification ($d \approx 0$, BoolQ). The gradient broadly tracks the coefficient of variation (CV) of the log-likelihood confidence proxy: LAMBADA (CV $=$ 0.75) $>$ ARC-Easy (0.54) $>$ HellaSwag (0.49) $>$ BoolQ (0.22). BoolQ's high-frequency short completions ("yes"/"no") yield elevated baseline log-likelihoods (mean $|\text{loglik}| = 3.04$ vs. 0.84 for LAMBADA) with similar absolute spread (IQR ≈ 0.9), compressing the relative variation and rendering the confidence margin unable to separate borderline-correct from robustly-correct examples.

Interpretation. The RetNet–HGRN2 anti-correlation may reflect different error modes specific to binary classification: when one architecture's recurrent state resolves a yes/no ambiguity through attention, the other architecture's resolution may be disrupted by the same attention intervention. This pattern warrants further investigation with additional binary classification benchmarks to determine whether it is BoolQ-specific or characteristic of two-choice tasks more broadly.

P WinoGrande Supplementary Analysis

WinoGrande ($N = 1{,}267$; reasoning, two-choice coreference resolution) shows high ambiguity ($\alpha \approx 0.41$ at h3-1) but mixed Jaccard significance: 10 of 24 architecture $\times$ attention-ratio pair combinations yield $p < .01$. Mean pairwise Jaccard ($\bar{J}_h = 0.110$, $\bar{J}_u = 0.103$) is comparable to the four main benchmarks, but the elevated null baseline from near-chance accuracy ($\sim$60%) reduces statistical power to detect above-chance overlap.

Near-chance accuracy. WinoGrande accuracy across hybrid configurations ranges from 59% to 62%, close to the two-choice chance level of 50%. At near-chance accuracy, a large fraction of examples are borderline—small perturbations (such as adding attention) can flip predictions in either direction, inflating the helped and hurt sets and consequently α. This structural property, shared with BoolQ ($\sim$59% accuracy, $\alpha \approx 0.45$), may partially explain the elevated α rather than task-intrinsic multiplicity. PIQA (73% accuracy, $\alpha = 0.14$) and HellaSwag (60% normalized accuracy, $\alpha = 0.15$) demonstrate that near-chance accuracy is not the sole driver, but we cannot fully exclude it as a con-

tributing factor for WinoGrande and BoolQ.

Table 22: WinoGrande predictive multiplicity at 1.3B (h3-1).

	N	α	$\bar{J}_h$	$\bar{J}_u$	Sig. pairs	ρ_α
WinoGrande	1267	.41	.110	.103	10/24	-0.9

α variation across ratios: 0.054 (> 0.03 threshold of main benchmarks).

Q 340M Per-Example Analysis

Q.1 LAMBADA

At 340M, LAMBADA per-example multiplicity closely matches the 1.3B pattern: $\alpha = 0.313$ (vs. 0.307 at 1.3B), with mean pairwise Jaccard $\bar{J} \approx 0.089$ (vs. 0.095). Per-example directionality is preserved: HGRN2 shows net positive effects at h3-1 ($+238$ examples), while GDN (-256) and RetNet (-106) show net negative effects, mirroring the 1.3B profile (HGRN2 $+153$, GDN -153, RetNet -69).

However, aggregate convergence is absent at 340M: $\rho = -0.50$ ($p = .45$), with non-monotonic σ across attention ratios ($0.025 \to 0.012 \to 0.018 \to 0.011 \to 0.016$). This contrasts sharply with 1.3B ($\rho = -1.0$, $p = .017$, monotonic σ decrease). The decoupling between per-example divergence and aggregate convergence thus manifests differently across scales: at 340M, per-example multiplicity is present but aggregate convergence is not, suggesting that convergence requires sufficient model capacity while multiplicity does not.

Within-benchmark α variation across attention ratios is also larger at 340M (0.047, from 0.313 at pure to 0.266 at h3-1) than at 1.3B (< 0.03; §4.2), suggesting that smaller models are more sensitive to the attention ratio in terms of per-example disagreement patterns.

Table 23: 340M vs. 1.3B LAMBADA comparison.

	340M	1.3B
α (pure)	.313	.307
$\bar{J}$ (h3-1)	.089	.095
ρ_σ	-0.50	-1.0
p	.45	.017
GDN net	-256	-153
HGRN2 net	$+238$	$+153$

Q.2 Cross-Benchmark Scale Comparison

Table 24 extends the scale comparison to all four main benchmarks. Two patterns emerge. First, α is systematically higher at 340M across all benchmarks, with the largest gap on ARC-Easy ($+8.4$pp) and the smallest on LAMBADA ($+0.6$pp). Second, pairwise Jaccard remains stable across scales ($\bar{J}_h = 0.08$–0.10 at both scales), confirming that the qualitative pattern of per-example multiplicity—which architectures affect which examples—is scale-robust even as its magnitude increases at smaller scale. Notably, the α inflation does not track accuracy loss: HellaSwag has the largest accuracy gap between scales yet the smallest $\Delta\alpha$ ($+1.9$pp), while ARC-Easy has a smaller accuracy gap but the largest $\Delta\alpha$ ($+8.4$pp), arguing against accuracy as the primary driver of scale-dependent multiplicity.

Table 24: 340M vs. 1.3B per-example multiplicity across four benchmarks. α: ambiguity at pure-recurrent configuration. Acc: mean accuracy across three architectures at pure-recurrent. $\bar{J}_h/\bar{J}_u$: mean pairwise Jaccard of helped/hurt sets. $\Delta\alpha$: 340M − 1.3B.

	340M				1.3B			
Benchmark	Acc	α	$\bar{J}_h$	$\bar{J}_u$	α	$\bar{J}_h$	$\bar{J}_u$	$\Delta\alpha$
PIQA	.663	.181	.090	.085	.143	.091	.094	$+.038$
HellaSwag†	.420	.167	.084	.083	.148	.095	.085	$+.019$
ARC-Easy	.594	.289	.098	.103	.205	.092	.099	$+.084$
LAMBADA	—	.313	.089	.090	.307	.095	.093	$+.006$

†`acc_norm`. LAMBADA accuracy omitted (evaluated in separate run).

Convergence is absent at 340M across all benchmarks (all $|\rho| \leq 0.4$, $p \geq .45$), in contrast to 1.3B where LAMBADA shows monotonic convergence. The absence of convergence at 340M extends beyond LAMBADA to HellaSwag ($\rho = 0.4$, $p = .52$), PIQA ($\rho = 0.3$, $p = .68$), and ARC-Easy ($\rho = -0.1$, $p = .95$).

R Subsample Sensitivity of α

To assess the reliability of α estimation at different sample sizes, we repeatedly subsample each benchmark (1,000 bootstrap replicates, seed 42) and compute the 95% CI width of α.

Table 25: α estimation stability: 95% CI width by subsample size.

Benchmark	$n = 100$	$n = 200$	$n = 500$	$n = 1000$	Full N
PIQA	.130	.095	.066	.045	.034
HellaSwag	.140	.100	.062	.044	.013
ARC-Easy	.160	.105	.072	.050	.032
LAMBADA	.180	.125	.082	.057	.025

At $n = 100$, CI width (0.13–0.18) spans the entire four-benchmark α range (0.14–0.30), rendering α estimation unreliable. This justifies excluding COPA ($n = 100$) from the per-example analysis.